

I develop methods for assessing structural systems under uncertainty: Bayesian inverse problems and identifiability from monitoring data, and the reliability of nonlinear structural systems.

POSITIONS

Independent research: Bayesian inverse problems and identifiability, Dresden	since 2025
R&D engineer, digital twins, MKP GmbH, Dresden	2025
Expert consultant, structural reliability and numerical analysis, engineering consultancy	2020 – 2024
Lecturer, master's programme ACCESS, TU Dresden	2019 – 2021
Senior research associate and project leader, Chair of Structural Design, TU Dresden	2014 – 2019
Research associate, Chair of Structural Design, TU Dresden	2008 – 2013
Doctoral candidate, Chair of Structural Design, TU Dresden	2006 – 2008
Lecturer and structural engineer, Homs University	2003 – 2005

SELECTED PUBLICATIONS

- Bakeer, T., Herbers, M., Marx, S. (2026).** Sensor informativeness, identifiability and uncertainty in Bayesian inverse problems for structural health monitoring. *Mechanical Systems and Signal Processing* 257, 114606.
- Bakeer, T. (2026).** Local information operators for spatial identifiability in distributed-parameter inverse problems in computational mechanics. *arXiv:2605.28601*.
- Bakeer, T., Herbers, M., Marx, S. (2026).** Information-guided Bayesian modelling for structural health monitoring. *Procedia Structural Integrity* (SMAR 2026), 1073–1080.
- Bakeer, T. (2023).** General partial safety factor theory for the assessment of the reliability of nonlinear structural systems. *Reliability Engineering & System Safety* 234, 109150.

STANDARDIZATION

CEN/TC 250: drafting of Annex X of EN 1990 (design assisted by numerical methods) in SC 10/WG 3; project teams for EN 1990 (nonlinear structural systems) and EN 1996-1-1.

DIN: committees on fundamental safety requirements (GruSiBau) and on earthquake safety of masonry.

EDUCATION

Habilitation (Dr.-Ing. habil.), Structural Engineering, TU Dresden	2017
Doctorate (Dr.-Ing.), Structural Engineering, TU Dresden	2009
Diploma in Structural Engineering, Homs University	2003

TEACHING AND SUPERVISION

Computational methods in engineering, master's programme ACCESS, TU Dresden	2011 – 2021
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Supervision of over 40 master's theses and project works; co-supervision of two doctorates.

ACADEMIC SERVICE

Member of the Joint Committee on Structural Safety (JCSS) and of GAMM. Reviewer for *Structural Safety*, *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Engineering Structures, Structures* and further journals.